ELEC3117 Final Report

Akane Opek (z5451214)
Yuhan Tang (z5531575)

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1 Executive Summary

The Medication Dispensing System, abbreviated MDS, is an assistive device for patients who regularly consume medication in the form of pills. The MDS aims to give these patients more autonomy and freedom by reminding them to take their medication at set times each day.

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2 Project Description - Yuhan

2.1 Introduction

2.2 Problem statement

The purpose of the Medicine Distribution System(MDS) is to inform patients or individuals who take supplements or prescriptions on a regular schedule to take their pills even while they are not at home. It is designed to be flexible to different environments, user needs, and schedules. The product guarantees that users will not face health problems anywhere, as the dispenser will automatically remind users to follow their prescription schedule. It aims to improve the quality of life of users by eliminating the need to remember to take medication or supplements on time during vacation or on weekends.

2.3 Needs Assessment

2.3.1 Who are you designing this product for?

Our target users include office workers, students, and anyone else who has to regularly and properly manage their medications or supplements not just at their homes.

2.3.2 Why do they need it?

It is exhausting for patients who are most likely away from home to pack medications and remember to take them on time, regardless of whether they are busy at work or on vacation.

2.3.3 What do they need and/or how will it be used?

The target audience needs a smart portable container that can set alarms to remind them to take their prescriptions on time. Meanwhile, it can monitor the temperature inside and the quantity of remaining medication. When the temperature is too high or the remaining medication is running low, the device should notify users to ensure both the quality and quantity of medications inside the medicine distribution system(MDS).

2.3.4 Where will it be used?

The product can be carried in a pocket and used anywhere, not just at home.

2.3.5 What features do they need?

The medicine distribution system(MDS) must at least be able to:

1. User-friendly interface to set alarms.
2. Provide both sound and light alerts to remind users when it is time to take their medicine.
3. Provide notifications to users when alarm triggered, time to refill, temperature is too high, etc.
4. Monitor the temperature inside the device.
5. Monitor the remaining quantity of medication.
6. Track the device if users accidentally leave it behind.
7. Real-time communication to ensure precise timing.
8. Update the time zone automatically for travel.
9. Software for user to send and receive data.

2.3.6 What are the issues of existing products/solutions?

The available choices on the current market have drawbacks of their own. For example, many of them are bulky in size and heavy to carry around, this restricts their usage in various environment. Besides, their greater size also leads to higher manufacturing costs.

2.3.7 Requirement Analysis

1. 3 x 3 doses of medication (3 days' supply, for 3 doses per day) should be safely stored in a container.
2. Produce an audio alarm that can be heard from both corners of a 10-square-meter space; the volume should be between 70 and 100 decibels.
3. Generate a visual alarm that flashes neon colors for at least 10 seconds each time. Light and portable casing that measuring 10x10x20cm in dimensions and weighing around 500g without medicine.
4. Only when the matching NFC reader is tapped onto the dispenser, the reliable electric solenoid lock is opened.
5. Continuously play an audio alarm that will beep if the corresponding tracker is out of range (e.g., ¿ 10m) to keep the user from losing the product.
6. Real Time Communication
7. A temperature sensor monitors the temperature inside the product to be less than 30°C(e.g., 17.06°C).
8. A pressure sensor detects pressure drops caused by medicine consumption(e.g., from 0.08hPa to 0.03 hPa).

2.4 Product Concept

We believe that a large group of people who are often not at home need to take medication on a regular schedule. For example, anyone who has to take their medicine whether working at an office, attending school, or going on vacation might benefit from the Medicine Distribution System(MDS). They would not have to carry along medicine bottles or set alarms on their phones if they brought the MDS along with them during the day. Users can simply configure when to dispense medicine and trigger alerts through its user-friendly interface on the device itself or software on their cellphone. It is less likely that a user would forget to take their medication at the alarm times if not only visual and auditory alerts included, along with notifications sent to their phone. Furthermore, the device monitors the amount of medication left in the container and the temperature inside, alerting users when it is going to run out or the temperature is too high, freeing them from having to check the quality and quantity of medications everyday by themselves.

To reduce the possibility of losing medication, wireless communication is also used to enable the dispenser to make a "beep" sound when it is out of range of its matching tag, which you can carry with you and place anywhere. Additionally, its cutting-edge technology enables it to automatically adjust its clock system when the user switches time zones, allowing the user to keep their medical routine. Finally, this dispenser's security lock may only be opened by permitted individuals. The MDS can provide a high level of protection for the user's medication and privacy.

2.5 Design Concept

Users can store their medications in 9 different compartments of the Medicine Distribution System (MDS). It consists of a speaker and a variety of LEDs to flash neon lights and play customizable audio alerts to let the user know when it is time to take their prescription. For the convenience of hard-of-hearing users, the volume of audio alerts can be changed by the user. Furthermore, customizable notifications, including pre-recorded messages, are supported by the MDS. In addition, the MDS will also display the user's dosing information on an LCD screen and send notifications to the user's cellphone to emphasize the alarm.

The MDS contains a number of different subsystems along with these fundamental functions. A pressure monitoring system that uses pressure sensors like the HX711 measures how many tablets are left. For better safety, NFC technology is employed in conjunction with an electric solenoid lock to communicate with the microcontroller. A tracking system uses wireless communication protocols(e.g. Bluetooth)to automatically update the time zone and a Bluetooth-enabled tag that beeps when the signal between them drops below a certain threshold.

A user authentication subsystem for better safety and privacy that uses NFC technology to work with an electric solenoid lock to communicate with the microcontroller.

2.6 USPs

1. The MDS's compact size and capacity enable the user to carry it in their pocket or bag wherever they go.
2. Time zones automatically updated so that the product prevents users from messing up their schedule to take medications, even if they are traveling overseas.
3. A corresponding tracker that allows user to track the location of MDS will trigger an alert when the dispenser is left out of a certain range.
4. Users do not have to keep an eye on how much medication is left since the pressure sensor will monitor the weight of the remaining medicine and inform the user to refill it when it is running low.
5. When the first alarm is set by user, the rest of alarms in a day can be automatically set by configuring how many times that user wants the alarm to be triggered in one day.
6.

3 Product Overview

3.1 High-Level Block Diagram

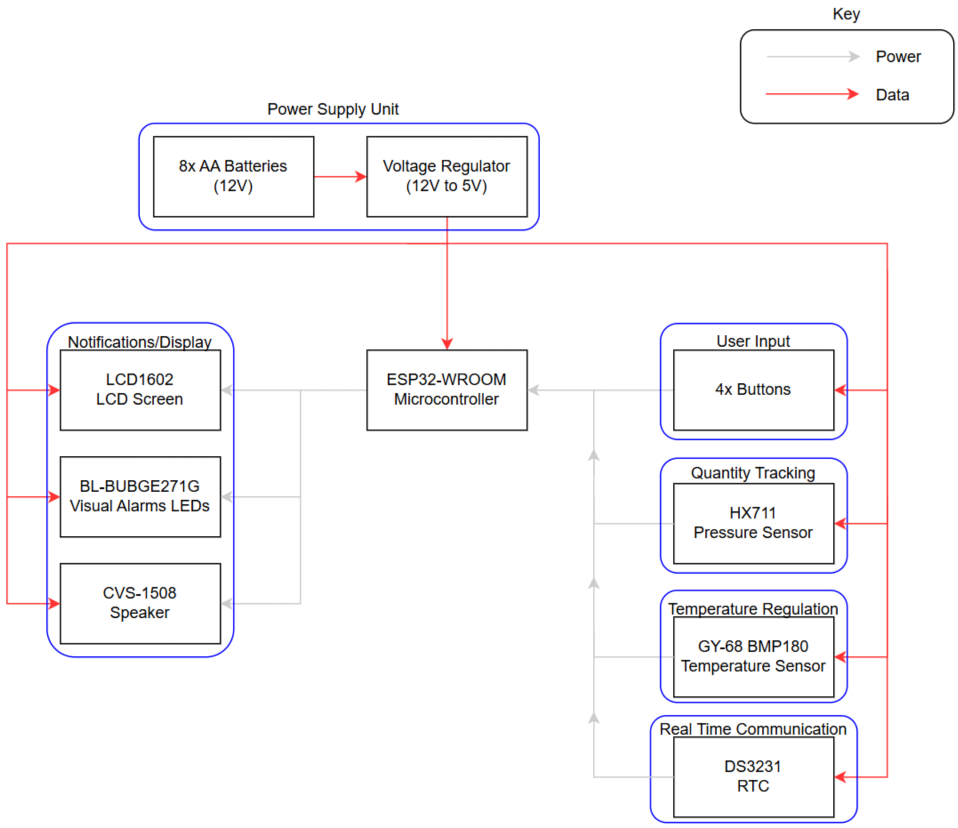

Figure 1: The prototype's High-Level Block Diagram

3.1.1 Power Supply

8 1.5V AA batteries are used as the main power supply. A DC-DC converter accepts 12V and outputs a regulated 3.3V for the input and output modules. The converter will ensure that over-current and over-voltage will not occur, so that the components will not be damaged. For the need of current time monitoring, the MDS lacks an on/off switch. Therefore, the MDS will near-always be a passive state consuming power.

3.1.2 Quantity Tracking

The micro-controller will initiate communication with the pressure sensor requesting information. The I2C protocol is followed between the micro-controller and pressure sensor, and so the pressure sensor plays passive part in this circuit. The micro-controller processes all data accordingly for display.

3.1.3 Temperature Regulation

Similar to the pressure sensor, the temperature sensor and micro-controller follows I2C protocol during communication.

3.1.4 Notification and Display Module

The alarm module consists of visual alarms (the LCD and LEDs) and an audio alarm (the speaker). The micro-controller provides a alert to when the alarm is triggered. At this time the LEDs will flash and toggle between on/off, the speaker will emit an audio alert, and the LCD will display 'Take Medicine Now!'. This variant of LCD uses I2C instead of SPI as its communication protocol to reduce the amount of pins - and therefore connections, which can reduce inductive coupling somewhat - needed.

3.1.5 Real Time Communication

Here, the DS3231 real time communication (RTC module) is used for time-keeping. A 32kHz temperature-compensated crystal oscillator is used in the DS3231, allowing for accurate real time values to be transmitted. Due to the nature of the DS3231, it is possible to retain the correct time even when the MDS loses power.

3.2 Principle of Operation

The MDS's main purpose is to alert users to take medication at set times while remaining portable. This is done by integrating a DS3231 for real-time communication, allowing the MDS to be aware of the current time. On the user's side, they are to set one or more alarms using four buttons as input. The micro-controller (ESP32) will continuously compare these values against the current time given from the DS3231, and will trigger an alarm when the values are identical.

When not the alarm triggered, the LCD is set to display several data values: the current time, the current ambient temperature, and the option to reset the set alarm time(s). First, by using several functions to get the current day, month, year, hour, minute, and seconds, the received DateTime object can be converted into a string for the LCD to then print out. Next, for the ambient temperature, the current temperature from the sensor to the micro-controller for the LCD to display. Third, the user can toggle the push buttons between different values to set an alarm and reboot the MDS. Debouncing features are built into the MDS's algorithm to maintain accurate input.

The micro-controller uses I2C communication protocols to interact with the peripheral sensors and LCD. For the purposes of the MDS, this is sufficient given the product's size and simplicity. In this case, I2C is able to offer more benefits to compensate for shortcomings against alternative such as SPI, namely the low power consumption and low pin count. Respectively, these two benefits allow for a longer shelf-life and smaller size so that the MDS will be more portable.

It can be noted that the micro-controller initiates all instance of communication between itself and its peripherals when transmitting data.

12V is supplied to the power supply unit, which will use a step-down converter to change this to 3.3V. 3.3V is supplied to the micro-controller, LCD, and speaker. The power supply unit will also regulate the supplied voltage and currents to prevent overdrawing.

4 Detailed Design - Yuhan

4.1 Design of Subsystems

The Medicine Distribution System(MDS) has multiple subsystems to meet user's needs as mentioned in previous sections.

4.1.1 Power Regulation System

The power regulation system is in charge of providing stable power to the whole system. It is supposed to produce a 3.3V rail for the microcontrolller from the 12V(8xAA) batteries, by using a step-down voltage converter.

4.1.2 User Input System

The user input system comprises four buttons for the user to key in the alarm time, integrate it together with the control system, user can easily set and cancel alarms.

4.1.3 Real-Time Communication System

The real-time communication system includes an RTC module, which allows the user to sync the current time and keep the time accurate even if the product is powered off. It works with the control system, once it matches with the alarm time, the notification system will be turned on to notify the user.

4.1.4 Notifications System

The notification subsystem consists of speaker, LEDs, LCD and user's phone. It is most widely used system that integrates with all the other subsystems and displays the corresponding data.

4.1.5 Quantity Monitoring System

The quantity monitoring subsystem uses pressure sensor like HX711 with a load cell to measure how many tablets or medications are left. When the pressure is below certain threshold, the notification system will emit alerts to warn user.

4.1.6 Temperature Monitoring System

The temperature monitoring system measures the current temperature inside the container using temperature sensor. When the temperature is too high, that might affect the quality and effectiveness of medications, the device will send notifications to warn users through notification system.

4.1.7 Control System

The state of the components for notification purpose like LCD, speaker, and LEDs can be controlled by using the ESP32-WROOM-32E chip, which is powered by 3.3V from the Power Supply PCB's output. The MCU can also receive the input data from other components(e.g., buttons, temperature sensor, pressure sensor) and use the data to determine the state of components used for notification.

4.1.8 User Authentication System

A user authentication subsystem for better safety and privacy that uses NFC technology to work with an electric solenoid lock to communicate with the microcontroller.

4.2 Technical Decisions

4.2.1 MCU

The MDS is programmed to operate on an ESP-WROOM-32 microcontroller. The main reasons for selecting ESP-WROOM-32 are its compact size and low power consumption. It will be in charge of communications with all the subsystems.

4.2.2 Power Regulation

The power supply subsystem uses the MC34063 IC because of its wide output range. In order to achieve the minimum inductance requirements for voltage switching in the power regulator, it must be used in combination with the SRN8040TA-820M as the inductor.

4.2.3 LEDs

The BL-BUBGE271G is selected as our LED, because of its ability to produce multiple colors and its through-hole design makes hand-soldering easier.

4.2.4 Speaker

CVS-1508 is chosen to be implemented as our speaker since it has a compact size, low power consumption, and water resistance.

4.2.5 LCD

The LCD1602 is selected due to its suitable screen size, as it will be used to display notifications of multiple subsystem results to user, such as the reminder of refilling.

4.2.6 Temperature Sensor

The temperature sensor we choose to implement in our product is GY-68 BMP180. The reasons for us to choose it are its small size, light weight and low power consumption.

4.2.7 RTC

The RTC module selected is DS3231 due to its long-term high accuracy, as it consists of a built-in temperature sensor to maintain time accuracy. Furthermore, it has a CR2032 coin cell holder, which provides a battery backup so that the time can be kept for years even if the main power of the product is off.

4.2.8 HX711

The HX711 is used as our pressure sensor to measure the accurate weight of the remaining medications.

Parameter	Microcontroller: ESP-WROOM-32	LCD: LCD1602	Speaker: CVS-1508	LED: BL-BUBGE271G	Temperature Sensor: GY-68 BMP180	RTC: DS3231	Pressure Sensor: HX711
Dimensions	18.10*25.6*3.2	80.0*36.0*12.0	15*7.5	2.54*30.8	3.6*3.8*0.93	38*22*14	35*21*3
Weight	2.3	30	2.0	1.0	0.01	8	0.01
Operating Voltage (V)	3.0	3.3	3.3	2.6	2.5	3.3	3.3
Current (mA)	80	20	190-250	30	0.003	NA	1.5
Power (mW)	240	66	300	80	0.01	NA	4.95
Operating Temperature (°C)	-40 to 85	-20 to 70	-0 to 55	-40 to 85	-40 to 85	0 to 40	-40 to 85
Accuracy	NA	NA	NA	0.01°C	NA	±2ppm	0.5g
Price (AUD)	12.95	13.5	6.55	0.49 (unit)	1.69	10	5.41

Table 1: Component table

5 Performance Analysis - Akane

5.1 Key Spefications

Based on user needs, the following product requirements have been created. The evaluated specifications are as follows.

Table 2: Table of the MDS's Key Specifications

The MDS must be able to keep the medication's integrity. All stored medication is to be kept intact in a range of different weather environments and temperatures.	Securely store 3 x 3 doses of medication (3 days worth of medication, for 3 doses per day) in a container that is: a.) water resistant, b.) able to retain the medication's and dispenser's shape/integrity in environments with 50 degree celsius ambient temperature, and c.) able to retain medication's and dispenser's shape/integrity from physical trauma, such as falls from 2 meters in height. The 3D printed enclosure was adequately water-repellent in light drizzles and was able to withstand a 2 meter fall, but may endure only up to 40 degrees Celsius ambient temperature before showing signs of warping.
The product muse be easily opened and accessible whilst remaining secure.	People who are disabled or with conditions such as Parkinson's or Carpal Tunnel Syndrome affecting mobility of the hands must be able to use and open the product.
The user interface must be intuitive and accessible, with distinguishable input points that can be differentiated from each other.	The MDS will use 4 evenly arranged buttons for input. The chosen button models are able to provide haptic feedback to allow the patient to recognize when an input is accepted.
Any audio alarm must be audible when triggered so that the alarm may be heard even outdoors.	Emit an audio alarm that is audible from opposite corners of a $10m^2$ room; the alarm should be within the 70 dB - 100 dB range The CVS-1508 has a speaker efficiency of 73.00 dBA. Amplifiers such as the LM
A visual alarm of sufficient luminosity and brightness must be seen each time the alarm is triggered.	Emit a visual alarm that flashes repeatedly in neon shades of color for at least 10 seconds each time. The LED flashed repeatedly for up to 30 seconds.
Portable and light, weighing 500g (without medication) and 10x10x20 cm in dimension.	The empty MDS has a total weight of roughly less than 400g, and dimensions of 15x4.5x4 cm.
The MDS must be secure enough to prevent spillage of medication and be tamper-proof to keep the medication's integrity.	The outer casing is designed to only open when sufficient force is applied to it, so that constant forces such as gravity will not be able to open the container. The 3D printed material is also resistant to wear and tear to provide a longer shelf life.

Measures should be present to minimize chances of the MDS being lost.	Consistently emit an audio alert that will beep whenever the corresponding tracker is out of range to prevent the user from losing the dispenser. A hole is drilled into a protruding corner of the outer casing for the user to hang the MDS onto a keychain or hook.
User must be able to remember, or be reminded, to purchase more medication when the stored amount is running low.	Pressure sensor measures the drops of pressure due to medicine consumption (e.g. drops from 0.08hPa to 0.03 hPa). Pressure Sensor HX711 was able to demonstrate changes on the force exerted on it, but failed to process changes as small as a pill's weight.
Battery life must be able to sustain the product's typical consumption for at least one week at a time.	The product's power supply unit is supplied 12V by 8 AA batteries, each one functional for 2mAH at 200mA.
End of Table	

5.2 Bill of Materials

The bill of materials is tabulated here. It can be noted that mass orders of several components will drastically reduce the individual price, leading to the final cost being reduced from what is expected.

Table 3: Bill of Materials for the production of one MDS

Description	Component	Bulk Quantity	Bulk Price ($)	Price per Unit ($)	Quantity	Total Price ($)
Micro-controller	ESP-WROOM-32	1300	15,327.00	12.95	1	12.95
LCD	LCD1602	N/A	N/A	13.5	1	13.5
Speaker	CVS1508	400	1,056.00	6.55	1	6.55
LED	BL-BUBGE271G	25000	4,325.00	0.49	3	1.47
Pressure Sensor	HX711	N/A	N/A	5.41	1	5.41
Temperature Sensor	GY-68 BMP180	N/A	N/A	1.69	1	1.69
RTC Module	DS3231	2500	25,727.00	10.00	1	10.00
Push Buttons	N/A, generic	5250	640.50	0.26	4	1.04
AA Battery	N/A, generic	100	29.95	1.11	8	8.88
12V - 3.3V Level shifter	MC34063	4000	2,740.00	1.52	1	1.52
Inductor	SRN8040TA-820M	10000	5,070.00	0.93	1	0.93

The total cost of the materials, disregarding PCB manufacturing costs and the PLA material for 3D printing the physical casing, is $63.94.
*For components where 'N/A' is listed under 'Bulk Quantity' and 'Bulk Price', bulk orders are not available or unable to be found on Mouser.

5.3 Performance Analysis

It should be noted that unforeseen challenges meant that we were unable to fulfill many of the subsystems' original purposes. The current MDS prototype only has the most basic functionality, which will be reflected in the performance analysis.

5.3.1 Quantity Tracking

The component HX711, which we chose partly due to financial constraints, was not sensitive enough to process changes in the magnitude of a pill or tablet's weight, and so it was unable to fulfill its intended purpose. However, it has a rated current of 1.5mA and rated voltage of 3.3V, for a power of 4.95mW. For one hour of use this consumes 4.95mAh of the available 16000mAh, which can easily be accommodated.

5.3.2 Temperature Regulation

The MDS lacks the space to properly implement temperature regulation. Instead, the values taken from the temperature sensor were constantly displayed, which is sufficient for this current prototype. We are given the rated values 0.003mA and 2.5V for a consumption of 0.01mW, which can be easily accommodated as well.

5.3.3 Notification and Display Module

Due to the model's constraints, the LEDs are limited in their luminosity, although sufficient for a prototype. The speaker may also require an amplifier for increased volume. On the other hand, the LCD was well suited for this purpose and did not pose a problem. Their combined power consumption is 446mW, which can still be accommodated.

5.3.4 Overall

As mentioned above, the current MDS prototype possesses the most basic functionality with a few additions. It is able to serve its intended purpose of alerting the user to ingest medication, but as of now offers little in way of uniqueness or marketability.

6 Business Plan - Yuhan

To estimate the potential business expenses and profits in the next 6-year,we have came up with a business plan for the product. The prototype of Medicine Distribution System(MDS) is build and planned to be released into the market within the following 3 months. We will spend about 3 months to collect user's feedback and improving our product to make it better satisfy customer demands until the product hits user satisfaction. After 6 months, the updated version will start to be manufactured to better satisfy client demands. Once the manufacturing is done, the product will be released to the market. We plan to sell our product to patients, NDIS-affiliated providers, and similar organizations to reach our target market.

6.1 Costs

The medicine distribution system's manufacturing expense can be divided into three parts: components, development and marketing.

6.1.1 The component cost

The MDS construction includes both mechanical and electrical components. Table 3 shows the estimated expenses and quantities for each electrical component. The prices for each component are provided on Mouser electronicsexcept for the microcontroller and pressure sensor, which are obtained from AliExpress. Mechanical component prices are ...

6.1.2 The development cost

Labour costs and installation equipment prices are the most significant components of development costs. As during the manufacturing, the workers only need to assemble the product and upload codes to the microcontroller, the development is quite simple and fast. Assume we have two engineers to develop the product,the time it takes to finish one product should be within two hours, and therefore the labour costs can be calculated to be $60 per product, with a cost of $30 per hour for each worker.

6.1.3 The marketing cost

As a supplier who provides products for sale, we do not need any registration or certification, and therefore we have no additional payments. Once our product is widespread among patients and clinics, we will make it available online and establish partnerships with trustworthy organizations like hospitals. Table 3 & 4 provides detailed cost estimates for all three categories, including unit pricing and quantity requirements for our product. The total cost is determined as AUD$1.. .

Category	Item	Quantity	Price	Total
Components	Electrical	1	$63.94	$63.94
	Mechanical	5	$3/hr	$15
Development	Labour	2	$30	$60
	Equipment	1	0	0
Marketing	Collaboration	NA	0	0

Table 4: Caption

Therefore, our total manufacturing costs is determined to be $138.94.

6.2 NPV

The selling price of our product is determined to be $165 according to our market analysis and estimate of manufacturing costs. Based on all the calculated costs so far and the NPV calculation parameters listed in Table 5, we can evaluate the net present value in the next 6 years as shown in Table 6.

Discount Factor P.A.	10%
Yearly Growth	10%
Selling Price	$165

Table 5: NPV Calculation Parameters

Year	0	1	2	3	4	5	6
Cost	$166,728	$183,401	$201,741	$221,915	$244,106	$268,517	$295,369
Unitsmanufactured	1200	1320	1452	1597	1757	1933	2126
Revenue	0	$217,800	$239,580	$263,538	$289,892	$318,881	$350,769
Future Value(Profit)	0	$34,399	$37,839	$41,623	$45,785	$50,364	$55,340
Present value	-$166,728	$31,272	$31,270	$31,271	$31,271	$	$
NPV	$	$	$	$	$	$	$

Table 6: NPV Calculations

7 Development Plan - Akane

Before being released into the market, the MDS prototype must undergo several rounds of review and redevelopment.

7.1 Work Breakdown Structure

Redevelopment will encompass the modification of existing electrical, physical, and software components to improve functionality. Further development of existing subsystems, namely the temperature regulation and quantity tracking functions, will also be carried out to better align with user needs alongside quality assurance tests. PCB design may also be performed to optimize the product's physical structure, electrical connections, and cost-effectiveness. This PCB will be designed and manufactured in tandem with the MDS's mechanical casing for physical coherence, allowing for optimal layouts.

Existing subsystems are to be modified, if not completely reworked, to meet higher standards of algorithmic complexity and security. All usage of external libraries in code should be reviewed for security vulnerabilities and dependencies or compatibility issues in all subsystems. Components in several subsystems will also need their suitability reviewed - particularly the quantity tracking subsystem, where the HX711 is not sensitive enough to process when the weight of one pill has been added or removed. Apart from the quantity tracking function, the temperature regulation subsystem requires review as well. For this subsystem to function as intended, it is necessary to add several hardware components, such as a relay module and heating/cooling elements. The combined size of these parts means that the MDS will no longer be portable, and so this subsystem will need to be reworked.

The PCB will need to be designed to accommodate the various sensors and modules (BMP180, DS3231, HX711). Given the difference in functions, their positioning will need to be reviewed. Multiple PCBs may be required to maximize each sensor or module's usage, and to minimize PCB trace inductance.

At the end of the development plan, tests are to be run to ensure that our product complies with applicable regulatory standards, such as Standard 3.2.2, Division 3 of the Australian Food Safety Practices and General Requirements. Tests may also be conducted to ensure that the product is ergonomic and accessible to people who are disabled, such as people with mobility conditions like Parkinson's, Arthritis, or Carpal Tunnel Syndrome.

Other changes to be made to the MDS includes altering the power supply module to accept 3 AAA batteries, instead of 12 AA cells (we used 12 because of a misunderstanding on project guidelines). The ESP32's I2S interface can also be used by adding an amplifier and potentiometer for volume control.

Link: https://www.foodstandards.gov.au/business/food-safety/food-packaging

1. **Review**
   (a) Review individual subsystems and modules to determine which areas require the most improvement to meet user requirements.
   (b) Test which subsystems and modules are working as intended, and check for dependencies or potential security risks in code due to the presence of external libraries or similar.
2. **PCB Design**
   (a) Redesign PCB to accommodate the input-output peripherals and the various sensors or modules.
   (b) Redesign the power PCB to become a boost up DC-DC converter instead of a step-down to decrease the MDS's size. The MDS should now be sustained on 3 AAA cells instead of 12 AA.
   (c) Check if EMC can be improved in the final design.

(d) Place PCB order with third parties such as JLCPCB or PCBWay. Check if PCBs are working as intended and redesign as necessary.

3. **Electrical Re-Design**

   (a) Incorporate the use of amplifiers such as the MAX98357A for increased maximum volume.

   (b) Incorporate a potentiometer or a volume control IC for volume adjustment.

4. **Physical Re-Design**

   (a) Increase the physical casing's width to be able to hold all components.

   (b) Add several PCB mounting stands as necessary.

   (c) Drill an optional keyring hole on the outer casing for patients who wish to attach the MDS to external objects.

   (d) Redesign for increased accessibility, ensuring that all patients will be able to use the MDS regardless of physical condition.

5. **Algorithm Re-Design**

   (a) Rewrite code to reduce dependency on external libraries and increase complexity.

   (b) Increase code readability by including more comments and removing redundant/repetitive sections, organizing code by subsystem and editing to have consistent style.

   (c) Reduce time complexity by streamlining code efficiency, introducing the use of interrupts.

6. **Quality Assurance Testing**

   (a) Create a criterion against which the product is to be compared.

   (b) Determine applicable quality standards and/or regulations and conduct relevant tests.

   (c) Redesign and perform tests as necessary, until satisfactory results are achieved.

As the MDS concerns medication, the review process must be made stringent to ensure that no health complications will arise from its usage. It may be worthwhile to partner with official medical organizations for the MDS's development.

7.2 Gantt Chart

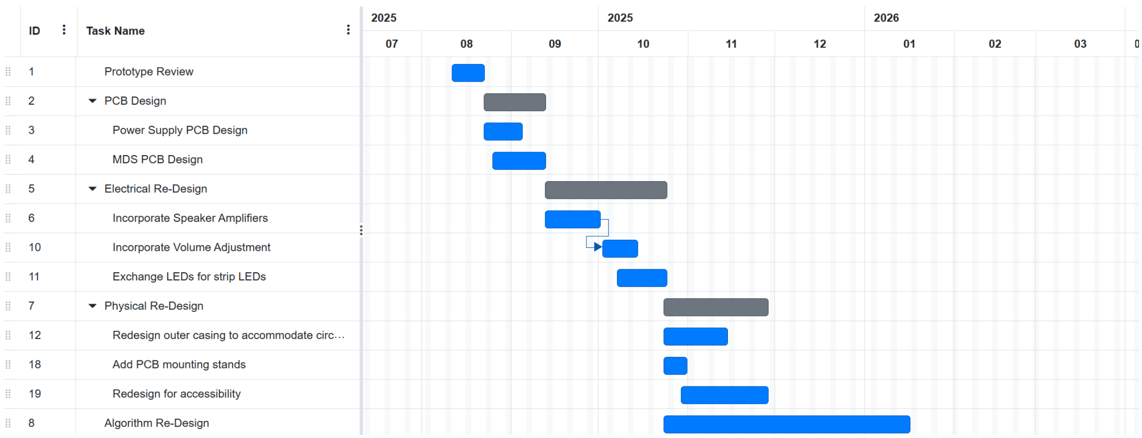

Figure 2: The prototype's High-Level Block Diagram

7.3 Risk Assessment

Table 7: Risk Assessment

Risk	Likelihood	Impact	Mitigation and Contingency
Technical Risks			
Major Design Flaw	Low	High	Repeatedly run extensive tests, creating a large tester pool to test the product in a wide range of conditions. Else, urgently recall product before it has been widely distributed, and undergo complete redesign.
Faulty Components	Low	Mid	Order components from multiple sources and run tests on each batch before use. Else, file a complaint with the manufacturer or distributor to try and get a refund or come to a compromise. Place express orders with different distributors and adjust remaining schedule accordingly.
Poor Soldering	Low	Mid	Only partner with reputable and experienced PCB manufacturers, or employ experienced engineers to solder the PCB or circuit in-house. Else, manually resolder each component or file a complaint with the manufacturer to come to a compromise.
MDS unable to keep medication integrity	Low	High	Create a double lining in the physical casing so that even if outer layer is compromised the inner layer will still be intact. Take reference from existing medication containers to receive guidance on how to keep medication integrity. Else, determine if a separate casing or lining can be added to the product to keep medication intact. If not, urgently recall the product and redesign.
Incorrect Interfacing	Low	Low	Design the product to have a larger room for error, and check each connection before being sent to manufacture. Else, rework connections and alter software to have the correct interfacing.
Overcurrent or Overvoltage	Low	High	Use components with suitable ratings, power components from the power supply module and integrate components such as a fuse or circuit breakers if possible. Else, replace components that broke as a result of overdrawing current or voltage.
Insufficient power supplied	Low	High	Simulate the product's circuit in LTSpice or similar software before being manufactures. Perform calculations to ensure that each component will receive its rated voltage or current. Else, increase the amount of power accepted into power supply module accordingly.
Business Risks			

Risk	Likelihood	Impact	Mitigation and Contingency
Market Volatility increases component pricing	Low	Low	Design with a large tolerance margin, so that components may be replaced with different manufacturers. Do not design with highly specific components. Else, redesign to use other components or sacrifice functionality is several subsystems to keep total cost low.
Market Demand for product sharply decreases	Low	High	Always keep monitoring the market and target audience, run marketing campaigns before product is launched and open pre-orders. Else, create a new USI that will appeal to the target audience and urgently redesign/remanufacture the product accordingly.
Competitor releases similar product during development	Mid	Mid	Continuously expand on the product's USIs and monitor competitors to determine the state of their product. Run marketing campaigns before product launch to build compnay reputation and trustworthiness; then open pre-orders. Else, create a new USI that will appeal to the target audience and urgently redesign/remanufacture the product accordingly.
Product fails to meet regulatory standards	Low	High	Always compare the product against applicable regulatory standards and run extensive testing. Else, redesign products to meet all standards.
Advancement in the medical field renders pill and tablet medication obsolete	low	Mid	Continuously monitor the medical field for related advancements, and adjust plans accordingly. Else, redesign the product so that it serves an adjacent purpose.
Management Risks			
Shipping Delays	Low	Mid	Place orders for components with different distributors, and place order as early as possible. Prioritize ordering components with longer shipping times or longer restocking times. Else, contact alternate manufacturers or distributors and pay the emergency fee depending on severity of delay.
Skill gap in engineering team	Low	High	Hire engineers with diverse skillsets to minimize gap, offer various training resources that will be able to minimize gap. Foster a work culture that will encourage engineers to keep learning and growing in their skills. Else, hire an external consultancy to close the gap and have engineers complete relevant training modules.

Risk	Likelihood	Impact	Mitigation and Contingency
Poor communication between RnD team and management	Low	High	Hold weekly or biweekly meetings to discuss the product's status as well as any other concern. Have all involved personnel complete training modules on communication. Provide templates for emails, reports and alternative communication forms. Else, hire external communication experts to facilitate communication.
Poor team harmony	Mid	Mid	Hire engineers who have demonstrated past skill in teamwork and leadership, and hire managers who possess complementary skill in managing engineers. Else, if poor harmony is due to one person's actions, remove that person.
End of Table			

8 Conclusion

9 References

10 Appendix